

Ana García del Molino, PhD

I engineer production AI systems that accelerate drug discovery — from reproducible deep learning pipelines to scalable agentic tooling for research scientists.
I set technical direction, build the infrastructure others rely on, and bring people along on the journey.

Spanish, English, French
a.g.delmolino@gmail.com
(+34) 675288835
garciadelmolino.github.io
GarciaDelMolino

Professional Experience

AstraZeneca, Barcelona
2024–Present
AI Engineer, Center for AI

AI Engineering and platform delivery for drug discovery.
Reproducible ML pipelines and production-ready AI applications.
Technical lead and community builder across R&D.

Technical lead and engineering counterpart for AI scientists, research teams, and IT to define, build, and operationalize reusable AI solutions for drug discovery.

Main Responsibilities:

- Technical lead for AI/ML applications and scientific software, from backend services and evaluation tooling to production deployment and observability.
- Ownership of Arete roadmap: design and delivery of the `cai-arete` package, Kubernetes and WandB CLI helpers, and multi-cluster compute strategy (Jade/Iron/Onyx).
- Ownership of CI/CD, release management, engineering standards, and shared developer tooling.
- Continuous enforcement of code quality: SonarQube, pytest coverage targets (25–60%), PR review process.
- Recruitment and management of external engineering resources: project scope, work allocation, and stakeholder alignment.
- Mentoring and 1:1 coaching of scientists and engineers — unblocking workflows, guiding contributors through reviews, and supporting career development.

AI Engineering for Drug Discovery: MLAB

- Delivered Library Generation and DeepScreening Hit Selection pipelines, empowering experimental scientists to run lead optimization experiments autonomously.
- Maintained foundational MLAB packages; raised test coverage and onboarded the team to feature-based branch development with isolated component environments (15+ Docker images in Harbor).

Technical Leadership & Design: Sophia

- Ensured production reliability: SSO support for large AD group memberships, CI/CD consolidation onto a unified deployment action, Phoenix/OTel tracing.
- Deployed DocEval and integrated it into the Sophia UI: one-click protocol evaluation against any benchmark, with token cost tracking.

Community & Knowledge Sharing: Software Development for AI CoP

- Led a company-wide agentic AI hackathon across 6 geographies and multiple business units; structured 14 cross-functional teams, 7 training sessions (40–60 attendees each).
- Took 40+ participants from near-zero proficiency to hands-on delivery in two weeks; post-event surveys showed ~80–95% adoption intent for core AZ tools (AI Gateway, RAGify, Agent Mesh, LangFlow).

Key takeaways and demonstrated skills:

- I have set the technical direction for engineering across CAI — not just implementing solutions, but defining the practices and standards the team now builds on.
- I have consistently bridged the gap between research scientists and production systems, translating scientific needs into reliable infrastructure without losing scientific flexibility.
- I have built and mobilised communities around AI adoption, taking people from zero to hands-on delivery.
- I am available at the 1:1 level — for the scientist who can't get their job to run, needs a review, or an updated workflow.
- I have grown in my ability to deliver through others — through leadership, planning, shared tooling, and clear documentation.

Agentic AI · Reproducible Pipelines · GitHub Actions · CI/CD · DevOps

Python · FastAPI · Kubernetes · LlamaIndex · Docker · Redis · OpenTelemetry · SonarQube

ByteDance - TikTok, Singapore
Sept 2020-2023
Computer Vision Scientist

Content Understanding for TikTok services.
ML Ops to support both Ecology and Business Integrity.
Mentoring of junior engineers and co-supervision of PhD candidates.

Technical counterpart for TikTok Live products and Business Integrity to define vision and deploy new features.

Main Responsibilities:

- Technical lead for code development and operations of online real-time services.
- Responsible for code development, release, and maintenance.
- Responsible for deployment high availability and fast response during outages.
- Point of contact for the infrastructure team to ensure smooth integration with new platform releases.
- Continuous communication with upstream teams to ensure data readiness in fast-paced environments
- Creation of reusable artifacts to enable fast deployments and new feature adoptions
- Model performance reporting with real-time dashboards for effective communication with stakeholders.

Product Ownership: content tagging; industry detection; risk assessment.

- Continuous improvement of the model design, training and benchmarking to support new business requirements.
- Pipeline design and maintenance for automated model iteration.
- Optimization of processes for data collection and labeling: data cleaning, deduplication and aggregation across multiple sources; automated collection of relevant training samples from the platform.

Full Cloud Environment:

- Datawarehouse workloads using hive (HSQL) similar to Redshift, Snowflake or Databricks.
- Big Data framework for ML pipelines and MLOps, similar to Glue and DataBricks.
- BI platforms for reporting and dashboarding, similar to Tableau, PowerBI or QuickSight.
- Streaming pipelines and content delivery with Kafka and Redis.
- ML development in Infrastructure as a Service (IaaS) and Platform as a Service (PaaS), similar to EMR, Sagemaker and Sagemaker Studio.

Multimodal Classification · MLOps · Batch and Real-Time Data Processing · Async Processes and Development · Data Analytics and Dashboarding

Python · TensorFlow · PyTorch · HSQL · PySpark · Golang · Redis · Kafka · JavaScript

BIGO Tech., Singapore
April 2019-Aug 2020
Sr. AI Algorithm Engineer

DCNNs to predict PCA parameters for graphic 3D modeling.

GANs for face expression manipulation.

DCNN-based autoencoders and U-Nets for image and video refinement and style transfer.

VAE · GAN · Face Landmark Detection and Tracking · Style Transfer · 3D Rendering

Python · TensorFlow · PyTorch · C# · Blender

gifs.com, Zurich
Summer 2017
Research Consultant

Use of a deep ranking model with personalizable input to customize the highlight detection in videos.

Action Classification · Recommendation Systems · Pattern Recognition · Action Understanding

Python · TensorFlow · DataFrames · SQL · Prototyping

I2R, A*STAR, Singapore
2014-March 2019
Ph.D. Candidate
Supervised by Joo-Hwee Lim

Visual Understanding and Personalization for an Optimal Recollection Experience

Modeling of lifelog photo-stream sequences using LSTM networks, to detect daily activities and learn associations between life events.

Use of conditional random fields and probabilistic inference for interactive video summarization with diversity and query-driven objectives.

Use of image quality analysis and semantic retrieval techniques for query-based lifelog summarization.

Object Classification · Activity Recognition · Facial Recognition · Transfer Learning · Spatio-Temporal Learning with LSTM · Episodic Memory

MATLAB · Caffe · Python · TensorFlow · C++ · Prototyping

National Inst. of Informatics, Tokyo
2013
Research Intern

Extension of the Bag of Words instance search technique by geometric coding and quantization error compensation.

Obtect Recognition and Retrieval · SURF & SIFT Descriptors · Hamming Embedding · Bag of Words

MATLAB · C++ · OpenCV

Education and Languages

PhD. in Cognitive Systems and Visual Computing

NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

School of Computer Science and Engineering
2015-2019

BSc. and MSc. in Telecomm Engineering

TelecommBCN (ETSETB)
2008-2013

Native
Native
Fluent
Proficient

Honors and Awards

Obra Social "La Caixa" 2014

Private fellowship to support postgraduate studies in North America and Asia-Pacific.

SINGA Scholarship 2015

Award given by A*STAR to international students to pursue their PhD at a Singapore University.

Publications

Wang Hao, Guosheng Lin, **Ana Garcia del Molino**, Anran Wang, Zehuan Yuan, Chunyan Miao, and Jiashi Feng. 2022. ManiCLIP: Multi-Attribute Face Manipulation from Text. *arXiv preprint arXiv:2210.00445.*

Qianli Xu, Fen Fang, **Ana Garcia del Molino**, Vigneshwaran Subbaraju, and Joo-Hwee Lim. 2021. Predicting Event Memorability from Contextual Visual Semantics. In *Advances in Neural Information Processing Systems.*

Ana Garcia del Molino, Joo-Hwee Lim, and Ah-Hwee Tan. 2018. Predicting Visual Context for Unsupervised Event Segmentation in Continuous Photo-streams. In *Proceedings of the 26th ACM international conference on Multimedia (MM '18).*

Ana Garcia del Molino and Michael Gygli. 2018. PHD-GIFs: Personalized Highlight Detection for Automatic GIF Creation. In *Proceedings of the 26th ACM international conference on Multimedia (MM '18).*

Ana Garcia del Molino, Xavier Boix, Joo-Hwee Lim, and Ah-Hwee Tan. 2017. Active Video Summarization: Customized Summaries via On-line Interaction with the User. In *AAAI Conference on Artificial Intelligence.*

Ana Garcia del Molino, Cheston Tan, Joo-Hwee Lim, and Ah-Hwee Tan. 2017. Summarization of Egocentric Videos: A Comprehensive Survey. In *IEEE Transactions on Human-Machine Systems, 47(1).*

Ana Garcia del Molino, Bappaditya Mandal, Jie Lin, Joo Hwee Lim, Vigneshwaran Subbaraju, Vijay Chandrasekhar. 2017. VC-I2R@ ImageCLEF2017: Ensemble of Deep Learned Features for Lifelog Video Summarization. In *CLEF working notes.*

Qianli Xu, Vigneshwaran Subbaraju, **Ana Garcia del Molino**, Jie Lin, Fen Fang, Joo-Hwee Lim, Liyuan Li, Vijay Chandrasekhar. 2017. Visualizing Personal Lifelog Data for Deeper Insights at the *NTCIR-13 Lifelog-2 Task.*

Jie Lin, **Ana Garcia del Molino**, Qianli Xu, Fen Fang, Vigneshwaran Subbaraju, Joo-Hwee Lim, Liyuan Li, Vijay Chandrasekhar. 2017. VCI2R at the *NTCIR-13 Lifelog-2 Lifelog Semantic Access Task.*

Ana Garcia del Molino, Qianli Xu, Joo-Hwee Lim. 2016. Describing lifelogs with convolutional neural networks: A comparative study. In *Proceedings of the first Workshop on Lifelogging Tools and Applications.*

Ana Garcia del Molino. 2016. First Person View Video Summarization Subject to the User Needs. In *Proceedings of the 24th ACM international conference on Multimedia (MM'16).*

Ana Garcia del Molino, Bappaditya Mandal, Liyuan Li, Lim Joo Hwee. 2015. Organizing and retrieving episodic memories from first person view. In *IEEE International Conference on Multimedia & Expo Workshops (ICMEW).*

(Check all my publications in my Scholar profile)